

GeoVLM-SceneReasoner

Geometry-Aware Visual Reasoning for Vision-Language Models

Huang Jiangzixuan | University of Hong Kong | github.com/Hjjj0918/GeoVLM-SceneReasoner

Project Goal. This project studies whether vision-language models (VLMs) can reliably answer real-image spatial reasoning questions, and whether explicit object-level geometry can improve their performance. The focus is not on training a new large model, but on building a lightweight, reproducible evaluation and enhancement pipeline that separates *perception failures* from *reasoning failures*.

Core Question. Can detection, segmentation, and monocular depth provide useful geometry signals for reasoning about object distance, relative position, support relations, occlusion, and physical size?

Pipeline.

Image	YOLO Detection	SAM2 Mask	Depth Anything V2	Object Geometry	VLM / LLM Reasoning
-------	-------------------	-----------	----------------------	-----------------	------------------------

- **Detection:** localize common scene objects and save per-image structured detections.
- **Segmentation:** use detection boxes as SAM2 prompts to obtain object masks.
- **Depth:** estimate monocular relative depth maps using Depth Anything V2.
- **Geometry:** summarize each object with bbox center, mask centroid, mask area, relative depth statistics, coarse position tags, and pairwise spatial relations.
- **Reasoning:** generate three comparable inputs: image-only VLM prompts, geometry-only LLM prompts, and GeoVLM prompts combining image and geometry.

Current Implementation. The current repository implements the full preprocessing and geometry-baseline path:

- real-image dataset layout, question schema, validation, and multi-view image renaming;
- YOLO detection, label normalization, detection visualization, SAM2 segmentation, mask visualization, and depth estimation;
- object-level geometry extraction and reasoning prompt generation;
- transparent geometry-only rule baseline and failure analysis;
- evaluation split generation that separates missing-target pipeline failures from geometry-available candidates.

Preliminary Local Benchmark Status. The current local multi-view test set contains 37 mobile-captured images and 111 question records. The automatic split reports 81 geometry-available candidate questions and 30 upstream pipeline-failure cases caused by missing target objects. Candidate questions still require manual mask/depth review before being treated as a clean evaluation subset.

Evaluation Design.

Track	Input	Purpose
Pure VLM	image + question	image-only spatial reasoning
Geometry-only LLM	object-level geometry + question	value of explicit geometry alone
GeoVLM	image + object-level geometry + question	whether geometry improves VLM reasoning

Future Plan.

- Build a manually reviewed clean subset by checking object masks, relative depth ordering, and question validity.
- Add VLM inference runners for pure VLM and GeoVLM settings, then compare against the geometry-only baseline.
- Expand question types to include left/right, front/back, occlusion, support, and physical-size reasoning.
- Evaluate on both self-captured images and public visual reasoning datasets to reduce bias from local image quality.
- Produce final comparison tables and failure-case analysis showing when geometry helps, when perception fails, and when reasoning remains difficult.

Positioning: A compact, non-robotics extension of geometry-aware multimodal reasoning, suitable for research on VLM visual reasoning, grounded perception, and lightweight benchmark construction.